

APPENDIX B: FULL WARM-START PARAMETER SWEEP RESULTS

λ_{L1}	λ_{TV}	λ_H	NRMSE ↓	SSIM ↑	STGE ↓	Temporal STGE ↓	Fourier χ^2 ↓
0	0	0	1.0068	0.4670	1.1488	1.0616	8.5964×10^{-7}
10^{-8}	0	0	1.0057	0.4683	1.1485	1.0613	8.5612×10^{-7}
10^{-7}	0	0	0.9962	0.4801	1.1461	1.0586	8.6257×10^{-7}
10^{-6}	0	0	0.9387	0.5299	1.1426	1.0467	4.2277×10^{-6}
10^{-5}	0	0	1.0000	0.5169	1.0000	1.0000	5.4243×10^{-5}
0	10^{-10}	0	1.4317	0.4079	1.1512	1.0419	5.9489×10^{-6}
0	10^{-9}	0	1.4320	0.4079	1.1507	1.0419	5.9491×10^{-6}
0	10^{-8}	0	1.4350	0.4078	1.1471	1.0426	5.9508×10^{-6}
0	10^{-7}	0	1.4529	0.4068	1.1317	1.0437	5.9690×10^{-6}
0	10^{-5}	0	1.8816	0.2735	1.1286	1.0648	8.1007×10^{-6}
10^{-8}	10^{-10}	0	1.4303	0.4092	1.1513	1.0419	5.9525×10^{-6}
10^{-8}	10^{-9}	0	1.4306	0.4092	1.1509	1.0420	5.9526×10^{-6}
10^{-8}	10^{-8}	0	1.4337	0.4091	1.1472	1.0426	5.9544×10^{-6}
10^{-8}	10^{-7}	0	1.4515	0.4080	1.1318	1.0437	5.9728×10^{-6}
10^{-7}	10^{-10}	0	1.4182	0.4203	1.1523	1.0421	6.0041×10^{-6}
10^{-7}	10^{-9}	0	1.4185	0.4203	1.1518	1.0422	6.0042×10^{-6}
10^{-7}	10^{-8}	0	1.4215	0.4202	1.1482	1.0428	6.0061×10^{-6}
10^{-7}	10^{-7}	0	1.4392	0.4193	1.1327	1.0440	6.0261×10^{-6}
10^{-7}	10^{-5}	0	1.8653	0.2990	1.1305	1.0657	8.2099×10^{-6}
10^{-6}	10^{-10}	0	1.3026	0.4964	1.1608	1.0427	8.1291×10^{-6}
10^{-6}	10^{-9}	0	1.3029	0.4965	1.1604	1.0428	8.1295×10^{-6}
10^{-6}	10^{-8}	0	1.3057	0.4965	1.1567	1.0435	8.1332×10^{-6}
10^{-5}	10^{-9}	0	0.8872	0.5293	1.2518	1.1461	4.8410×10^{-5}
10^{-5}	10^{-7}	0	0.8944	0.5290	1.2582	1.1691	4.8475×10^{-5}
10^{-5}	10^{-5}	0	0.9373	0.5192	1.7304	1.7218	5.3521×10^{-5}
0	0	10^{-7}	1.7494	0.4818	1.1591	0.9764	1.4267×10^{-5}
0	0	10^{-5}	1.7414	0.4812	1.0712	0.9755	1.4331×10^{-5}
10^{-7}	0	10^{-7}	1.7473	0.4886	1.1596	0.9762	1.4343×10^{-5}
10^{-7}	0	10^{-5}	1.7392	0.4885	1.0715	0.9753	1.4405×10^{-5}
10^{-5}	0	10^{-7}	1.5970	0.5842	1.2216	0.9601	2.2528×10^{-5}
10^{-5}	0	10^{-5}	1.5841	0.5816	1.1089	0.9601	2.2495×10^{-5}
0	10^{-7}	10^{-7}	1.7659	0.4868	1.1633	0.9774	1.4677×10^{-5}
0	10^{-7}	10^{-5}	1.7563	0.4869	1.0764	0.9764	1.4725×10^{-5}
0	10^{-5}	10^{-7}	1.8773	0.4618	1.1474	0.9772	1.5044×10^{-5}
0	10^{-5}	10^{-5}	1.8689	0.4606	1.0954	0.9774	1.5057×10^{-5}
10^{-7}	10^{-7}	10^{-7}	1.7640	0.4933	1.1638	0.9773	1.4746×10^{-5}
10^{-7}	10^{-7}	10^{-5}	1.7543	0.4940	1.0767	0.9763	1.4793×10^{-5}
10^{-7}	10^{-5}	10^{-7}	1.8755	0.4729	1.1479	0.9771	1.5097×10^{-5}
10^{-7}	10^{-5}	10^{-5}	1.8670	0.4721	1.0959	0.9773	1.5110×10^{-5}
10^{-5}	10^{-7}	10^{-7}	1.6289	0.5852	1.2214	0.9637	2.2081×10^{-5}
10^{-5}	10^{-7}	10^{-5}	1.6152	0.5829	1.1125	0.9634	2.2054×10^{-5}
10^{-5}	10^{-5}	10^{-7}	1.7420	0.5800	1.2102	0.9643	2.2460×10^{-5}
10^{-5}	10^{-5}	10^{-5}	1.7295	0.5777	1.1446	0.9651	2.2417×10^{-5}

Complete warm-start regularization sweep using the radial-template initialization and 200 iterations. The table combines all unique parameter configurations evaluated across the L1, TV, and Hessian sweeps. The arrow next to each metric indicates the numerical direction of what is ‘better’ for that metric. The bold L1-only configuration, $\lambda_{L1} = 10^{-6}$, $\lambda_{TV} = 0$, and $\lambda_H = 0$, was selected as the principal warm start because it provides the strongest overall balance between NRMSE, SSIM, STGE, temporal STGE, and Fourier consistency. The absolute minimum NRMSE is achieved with $\lambda_{L1} = 10^{-5}$ and $\lambda_{TV} = 10^{-9}$, but this configuration substantially degrades both STGE and Fourier-domain consistency. The $\lambda_{L1} = 10^{-5}$, zero-TV, zero-Hessian case collapsed to an all-zero or nearly all-zero solution.